1. N/A
2. **Explain the difference between method declaration and method body.**

Method Declaration - The first line of a method, containing the method name, access level, return type, and parameters.

“What it is”

Method body - The statements that implement a method.

“What it does”

3. **What type of keyword is used to change the access level of a method?**

Access modifier - A keyword in the declaration of a method that determines the access level of a method.

Examples.
- Public (method can be accessed from anywhere)
- Private (method can be accessed only within the same class)

4. **What is another word used for describing the access level of a method?**

Visibility - the access level of a method, indicates who can see or use the method.

5. **Explain the scope of each of the variables in the code below:**

Var1
- Declared in main(): int var1;
- Scope: From the declaration until the end of main()
- Accessible: anywhere inside the main method
- Not accessible: in method1() or outside the class

Var2
- Declared in the *for* loop of *main*: *for (int var2 = 0;* …
- Scope: only inside the *for* loop where it is declared
- Accessible: only within the { } of that loop
- Not accessible: Outside the loop or in *method1()*

Var3
- Declared in the *for* loop of *method1()*: *for (int var4 = 0;* …)
- Scope: only inside the *for* loop in *method1()*
- Accessible: only within the { } of that loop
- Not accessible: outside the loop or in *main()*

Var4
- Declared in the *for* loop of *method1()*: *for (int var4 = 0;* …)
- Scope: only inside the *for* loop in *method1()*
- Accessible: only within the { } of that loop
- Not accessible: Outside the loop or in *main()*

6. **Write a method declaration for each of the following descriptions:**
   a) Public static int getVowles(String input)
   b) public static int extractDigit(int number)
   c) public static String insertString(String text, int position)

7. **How does the compiler distinguish one method from another?**
The compiler uses method name + parameter list to identify and distinguish methods.
Can two methods in the same class have the same name?
Yes, two methods in the same class can have the same name, but only if they have a different method signature. This is called method overloading. Method overloading is writing more than one method of the same name in a class.

8. **What is the return statement used for?**
The return statement is used to send a value back from a method to the part of the program that called it. Or exit the method immediately, stopping any further code in that method from running.
How many values can a return statement send back to the calling statement?
1 value
How is the declaration of a method returning a value different from the declaration of a method that does not return a value?
Return type tells whether the method gives something back or not. A method declaration that does not return a value can only be used to exit the program early, whereas one that returns a value, will return a value.

9. **FInd and explain the error in the code below:**
Placement of *main*
All methods must be inside a class. In this code *main* appears outside the class *MethodCallExample,* which is not allowed in java. This will result in the code not compiling. Meaning java refuses to turn the code into a program because something is written incorrectly.

11. Determining if the following are true or false. Explain.
   a) True, Procedural abstraction is the process of dividing a program into smaller, manageable methods that perform specific tasks
   b) False, a method call is when you invoke a method, not declare it. The declaration defines the method, the call uses it.
   c) False, a void method does not return any value. It can use *return*; to exit early but does not return data.

d) False, Access modifiers control visibility not the return type. The return type is declared separately.
e) False, static makes a method a class method. The return type is independent.
f) False, Method parameters are enclosed by parentheses not braces.
g) False, Local variables are only accessible within the method or block where they are declared.
h) True, for primitive types, changing the parameter inside the method does not affect the original argument. For objects, contents can be changed inside the method.
i) False, method overloading means having multiple methods with the same name but different signatures.
j) True, methods with a return type use *return* to send the result back to the caller.
k) False, a precondition is a condition that must be trie before the method is called.
l) False, a postcondition states what is true after the method executes, not the specific steps of how it works.